

Keywords: single-image generation • diffusion models • patch statistics • training-free

Efficient and Training-Free Single-Image Diffusion Models

Closed-Form Patch Denoising Replaces Hours of Training with Seconds of Inference—CVPR 2026 Highlight

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arXiv:2606.04299 • CVPR 2026 Highlight • Submitted June 2026

Single-image generative models no longer need to be trained. By treating a reference image as an empirical patch distribution and computing the score function analytically in closed form, a new method from the University of Toronto generates diverse, high-quality samples in seconds—300× faster than trained alternatives—while matching or exceeding their quality on standard benchmarks.

AT A GLANCE

Problem: Generating novel images that match the internal patch statistics of a single reference image currently requires hours of neural network training.

Why it matters: Training-based single-image methods are too slow for interactive editing, personalised synthesis, or low-resource scenarios; a training-free alternative removes the bottleneck entirely.

Method: Model the image as a finite empirical distribution over patches at multiple scales; compute the diffusion score function in closed form via Gaussian kernel regression over that patch set.

Result: State-of-the-art SIFID on most textures, 300× speed-up over SinDDPM, and support for text-guided stylisation and retargeting without any fine-tuning.

Evidence: CVPR 2026 Highlight; evaluated against SinDDPM, SinFusion, SSGAN, ConSinGAN on standard single-image benchmarks.

The Big Picture

Single-image generation (SIG) asks: given one photograph or texture, synthesise new images that share its internal visual statistics. Applications span texture synthesis, artistic style transfer, image retargeting, and data augmentation for rare visual classes. The task is intrinsically self-supervised—the only training signal is the image

itself.

The dominant recent approach trains a small diffusion model on the patches extracted from a single image (SinDDPM, SinFusion). While effective, even these small networks require hours of GPU time per reference image—a steep price when the target is a single photo. The underlying cost is neural network training, which is needed because the score function is otherwise intractable for a general data distribution.

This paper observes that for a single image the score function *is* tractable: the image contains only finitely many patches, so the score at any noisy point is exactly the posterior mean over that finite set—a Nadaraya-Watson kernel regression with a Gaussian kernel, computable in closed form. Embedding this closed-form denoiser inside a standard diffusion sampling loop yields a complete, training-free generative model that runs in seconds on a commodity GPU.

Why Existing Approaches Fall Short

Neural single-image diffusion models (SinDDPM, SinFusion) require training a U-Net on the reference image from scratch. Training takes 1–8 hours depending on image resolution, making interactive use impractical. Moreover, these networks are *not* reusable: each new reference image requires fresh training.

Score function engineering approaches (e.g., SinGAN) use a GAN pyramid but suffer from training instability and do not support the flexible text-guided generation that diffusion models enable.

The fundamental gap is the absence of a principled, efficient score estimator that does not require parametric training.

Core Mechanism

For a single image I , extract patches $\mathcal{P} = \{x_1, \dots, x_N\}$ at multiple scales. For a noisy patch $y = x^* + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$, the optimal minimum-MSE denoiser is the posterior mean:

$$D_\sigma(y) = \mathbb{E}[x | y] = \frac{\sum_{i=1}^N x_i e^{-\|y-x_i\|^2/2\sigma^2}}{\sum_{i=1}^N e^{-\|y-x_i\|^2/2\sigma^2}}.$$

This is Gaussian kernel regression over the patch set—no neural network required. The score function follows from Tweedie’s formula:

$$\nabla_y \log p_\sigma(y) = \frac{D_\sigma(y) - y}{\sigma^2}.$$

Technical Formulation

Patch dataset. For resolution $h \times w$ and patch size $p \times p$, the patch set $\mathcal{P}^\ell$ at scale ℓ has $N^\ell = (h_\ell - p + 1)(w_\ell - p + 1)$ elements, each of dimension $d = p^2 \cdot C$ (with C channels).

Optimal denoiser. Under the empirical prior $p(x) = \frac{1}{N} \sum_{i=1}^N \delta(x - x_i)$, the posterior given noisy observation y is:

$$p(x | y) \propto \mathcal{N}(y; x, \sigma^2 I) \cdot p(x) = \sum_{i=1}^N w_i(y) \delta(x - x_i),$$

$$w_i(y, \sigma) \propto \exp\left(-\frac{\|y - x_i\|^2}{2\sigma^2}\right).$$

The posterior mean $D_\sigma(y) = \sum_i w_i x_i$ is the closed-form denoiser.

Computational cost. Each denoising step costs $O(Nd)$ inner products. For $p = 8$, $C = 3$ and a 512×512 image, $N \approx 2.5 \times 10^5$ and $d = 192$, giving $\approx 5 \times 10^7$ operations—substantially cheaper than a forward pass of even a small U-Net.

Multi-scale generation. Coarse-to-fine: generate at the lowest resolution first using $\mathcal{P}^1$, upsample, and condition the next-scale denoiser on the upsampled result via score guidance:

$$\tilde{\nabla}_y \log p_\sigma^\ell(y) = \nabla_y \log p_\sigma^\ell(y) + \omega \nabla_y \log p_\sigma^{\ell-1}(y_\downarrow),$$

where $y_\downarrow$ is y downsampled to scale $\ell-1$ and ω is a guidance weight.

Learning or Inference Procedure

No training is performed. At inference:

1. Extract patch sets $\{\mathcal{P}^\ell\}_{\ell=1}^L$ from I .
2. Initialise $x_T^1 \sim \mathcal{N}(0, I)$ at the coarsest scale.
3. Run DDPM reverse process using D_σ as the denoiser (instead of a neural network) for T steps.
4. Upsample, condition, and repeat for finer scales.
5. Output x_0^L as the generated image.

Total wall-clock time: < 10 seconds on a single GPU for a 256×256 output, vs. > 3 hours for SinDDPM.

What the Guarantee Says

The paper proves that D_σ is the *Bayes-optimal* denoiser under the empirical patch prior: no estimator achieves lower MSE for that prior. As $N \rightarrow \infty$ (more patches, i.e., larger image), the empirical prior concentrates on the true patch distribution and D_σ converges to the score function of the true distribution—so the method is asymptotically exact.

Experimental Findings

On the Oxford Textures and USC-SIPI image datasets:

- Lower SIFID (Single-Image FID) than SinDDPM on 8 of 10 tested textures.

- 300× faster at inference (seconds vs. hours).
- Competitive FID on random sample diversity.
- Text-guided stylisation (using CLIP guidance on the denoiser score) produces coherent results with no fine-tuning.
- Image retargeting (change aspect ratio while preserving texture) outperforms SinGAN in perceptual quality.

Ablations and Interpretation

Patch size p : $p = 8$ gives the best SIFID/diversity trade-off; larger patches lose texture variety (fewer distinct patches), smaller patches lose global coherence.

Number of scales L : $L = 3$ is sufficient; $L = 1$ generates texturally correct but spatially incoherent images.

Guidance weight ω : Higher ω produces images more faithful to the reference layout; lower ω increases diversity.

Reference

Haojun Qiu, Kiriakos N. Kutulakos, David B. Lindell. **Efficient and Training-Free Single-Image Diffusion Models**. CVPR 2026. arXiv:2606.04299. <https://arxiv.org/abs/2606.04299>